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Review Article

Space-time cluster detection techniques for infectious diseases: A systematic review

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A R T I C L E I N F O

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A B S T R A C T

Background: Public health organizations have increasingly harnessed geospatial technologies for disease surveillance, health services allocation, and targeting place-based health promotion initiatives.
Methods: We conducted a systematic review around the theme of space-time clustering detection techniques for infectious diseases using PubMed, Web of Science, and Scopus. Two reviewers independently determined inclusion and exclusion.
Results: Of 2,887 articles identified, 354 studies met inclusion criteria, the majority of which were application papers. Studies of airborne diseases were dominant, followed by vector-borne diseases. Most research used aggregated data instead of point data, and a significant proportion of articles used a repetition of a spatial clustering method, instead of using a "true" space-time detection approach, potentially leading to the detection of false positives. Noticeably, most articles did not make their data available, limiting replicability.
Conclusion: This review underlines recent trends in the application of space-time clustering methods to the field of infectious disease, with a rapid increase during the COVID-19 pandemic.

1. Introduction

Infectious diseases, also known as communicable diseases, are defined as "diseases caused by pathogenic microorganisms, such as bacteria, viruses, parasites or fungi; the diseases can be spread, directly or indirectly, from one person to another" (WHO, n.d.). The spread or transmission route could be through direct and/or indirect contact, airborne, waterborne or foodborne, vector-borne, and the environment. For instance, the transmission of Coronavirus disease 2019 (COVID-19) among humans includes both direct and indirect contacts (such as contaminated surfaces) and airborne aerosol/droplet routes; the latter is considered the dominant transmission mechanism for this disease (Zhang et al., 2020).

Timely surveillance is an essential epidemiological component to describe the ongoing dynamics of a disease, monitor trends, and detect outbreaks and new pathogens (Murray and Cohen, 2017). Three key factors are critical when conducting space-time surveillance of infectious diseases: (1) the time when the disease occurs, (2) the geographic location where cases are reported, and (3) the segments of the population who get infected. Documenting each of these factors is challenging when data is uncertain or incomplete. For instance, individuals may be infected days before a test reveals positivity due to the incubation period. Or individuals may be asymptomatic, increasing uncertainty in the number of daily cases. Reported cases may originate from another region than where the test for the disease was conducted at. Despite those issues and uncertainties, effective prevention methods, monitoring and early detection of outbreaks, and controls and treatments aimed to stop or slow down the spread of infection have improved (Smith et al., 2014). Among the technologies that contribute to infectious diseases surveillance are geospatial technologies, which are particularly suited to capture the geographic complexity of infectious diseases (Kirby et al., 2017).

A geographic information system (GIS) is a computer system for creating, managing, analyzing, and displaying geospatial data, a valuable and practical tool for monitoring infectious disease research (Cromley, 2003; Eisen and Eisen, 2014; Kirby et al., 2017). As early as 1854, John Snow, the father of modern epidemiology, identified a cluster of cholera cases in London around a public water pump, which became the source of the outbreak due to contaminated water found in that pump (Newsom, 2006). Back at that time, cholera was a dangerous

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infectious disease which would mainly spread through contaminated water. This example has been used in many textbooks to showcase how a simple map of deaths could uncover the distribution of an infectious disease. Obviously, the display of data on the map is not the most important tool in epidemiology (Kistemann et al., 2002); other geospatial functionality can be used for geographic data collection, management, and spatial analysis, such as geocoding, overlay, interpolation and proximity analysis to name a few. GIS can be used in concert with detection tools to monitor and respond to health issues, ultimately assisting health professionals in identifying cases, spatial trends, disease clusters, and correlation with other spatial data (Carroll et al., 2014; Delmelle et al., 2011; Delmelle et al., 2015). As a vital component of disease surveillance, cluster detection has the ability to identify high-risk areas, and it can facilitate the investigation of the spread of infectious diseases (Aamodt et al., 2006).

Space-time cluster detection methods play a pivotal role in infectious disease spread, as the key to understanding the diffusion of infectious diseases is to uncover the dynamic process of spreading patterns with the investigation of time, place, and person at the same time (Rogers and Randolph, 2003), which is the major difference from other diseases. Epidemic models are used to measure the dynamics of an infectious disease through different groups of the population during its spread. However, infectious diseases are not merely related to infected populations and populations at risk, as the risks of infection and transmission are caused by a myriad of covariates, such as demographics, socioeconomic factors, and environmental characteristics. These conditions can act as confounding factors leading to variations in the risk of infection and transmission (Delmelle et al., 2016; McMichael, 2004; Taylor et al., 2001; Weiss and McMichael, 2004; Wichmann et al., 2007). In addition, as evidenced by the COVID-19 pandemic, interactions among individuals can accelerate the spread of disease, adding another layer of complexity to the analysis of infectious diseases.

Before we introduce cluster detection techniques, it is fundamental to highlight conceptual differences between clusters, outbreaks, and hotspots. These terms are often used interchangeably in diseases surveillance. Knox (1989) provided a non-mathematical definition of a *cluster* as “a geographically bounded group of occurrences of sufficient size and concentration to be unlikely to have occurred by chance.” As to *hotspot*, Lessler et al. (2017) summarized three distinct types in epidemiology and suggested alternative terms such as *transmission hotspot* (elevated transmission efficiency), emergence hotspots (a high frequency of emergence or re-emergence of diseases), and *burden hotspot* (elevated disease incidence or prevalence or a geographic cluster of cases). Among these three types, the definition of *burden hotspots* is similar to clusters, as it was defined as “a geographic cluster of cases.” Farrington and Beale (1998) described an *outbreak* as the increment in the number of cases beyond expected levels. According to those definitions above, an outbreak refers to the status of unexpectedly elevating infected cases in areas during a particular time. During an outbreak, hotspots and clusters are areas with elevated incidence or prevalence or sufficient concentration of the disease, and these two terms are commonly interchangeable. For example, the current pandemic is an outbreak of COVID-19, while hotspots and clusters of COVID-19 cases are dynamic in both spatial and temporal dimensions. This review is primarily concerned with “clusters,” particularly space-time clusters, as they also imply statistical significance that hotspots do not provide.

Although forecasting is a necessary tool in public health responses, the focus of this review is to understand the ’nowcasting,’ the current situation, and the presence of clusters. Thus, we mainly focused on space-time cluster detection methods using statistical test. This review does not cover the large literature around Bayesian model-based detection approaches, which can be superior to testing based methods, such as Moran’s I and SaTScan (see Lawson, 2016; Lawson and Denison, 2002).

1.1. Techniques for cluster detection

Several spatial statistics methods have been developed to detect areas of elevated risks (see Fig. 1), and these can broadly be classified into three categories based on their mechanisms: (1) distance-based methods that measure separation among cases, such as the *K* function; (2) area-based methods that analyze cases within subset regions of a study area, such as spatial autocorrelation and scan statistics and (3) continuous methods that estimate a risk surface, such as kernel density analysis.

The two most common data types used in health surveillance are disaggregated data (e.g., point) that contain spatial coordinates for each case and areal or aggregated data that aggregate cases for a specific region (e.g., postal code) during a given time. Aggregated data are typically more available and less sensitive to privacy issues (Olson et al., 2006).

Some cluster detection techniques have been extended to deal with temporal information in two different ways: (1) the statistical method is repeated over different time periods; or a (2) a more robust approach that explicitly takes space and time into account can be developed. These cluster detection methods are typically implemented as part of specific standalone software such as CrimeStat (Levine, 2013), GeoDa (Anselin et al., 2010), SaTScan (Kulldorff, 2010), SpaceStat and ClusterSeer from Biomedware, or codes in opensource programming platforms using R (Bivand et al., 2008; Gómez-Rubio et al., 2005; Moraga, 2017), python (Rey and Anselin, 2010), etc. However, those methods suffer from poor visualization, especially in space and time.

1.2. Point data

Techniques to evaluate clustering among point data can be categorized into methods with first-order and second-order variation or properties. First-order variation indicates that the point process varies over space because of spatial variation in the mean (for example, cholera cases may vary depending on contaminated water distribution). In contrast, second-order variation describes that the variation in a spatial process is associated with spatial dependency (Gatrell et al., 1996; Rogerson and Yamada, 2008). For example, areas with high community transmission could result in high rates of COVID-19. Among first-order variation techniques, we describe the Kernel Density Estimation (KDE) and interpolation methods, including Inverse Distance Weighted and Kriging; and among second order methods, we explain the nearest neighbor statistic and the *K*-function.

The first-order variation techniques can map the variation of certain event, such as the patterns of the disease. One such techniques is KDE, which calculates the intensity of observed points over the study region. The study region is divided into a grid of square cells, and the intensity at each cell is estimated using the Kernel function. This weighted distance function measures the intensity between the cell’s centroid to all events within a predefined bandwidth. The result of the KDE for a given cell is the sum of the intensities within the bandwidth. As the choice of bandwidth is arbitrary, KDE results are subject to a tradeoff between bias due to a considerable bandwidth and uncertainty due to a small bandwidth (Rogerson and Yamada, 2008).

Although less prevalent in infectious diseases, interpolation methods such as Inverse Distance Weighted or Kriging can monitor the spatial variation of contagious diseases, but they require an attribute (Z-value). Typically, data are aggregated at an areal level (e.g., count or rate of events in a zip code), however the data are generally reduced to a point, such as the centroid of that area. Interpolation methods will then use these points and attribute as input to generate a continuous surface. One crucial assumption of interpolation methods is that the spatial correlation structure is spatially constant, suitable for environmental variables such as water or air pollution, but not infectious cases or rates (Pfeiffer et al., 2008).

Unlike those first-order techniques that only map disease patterns,

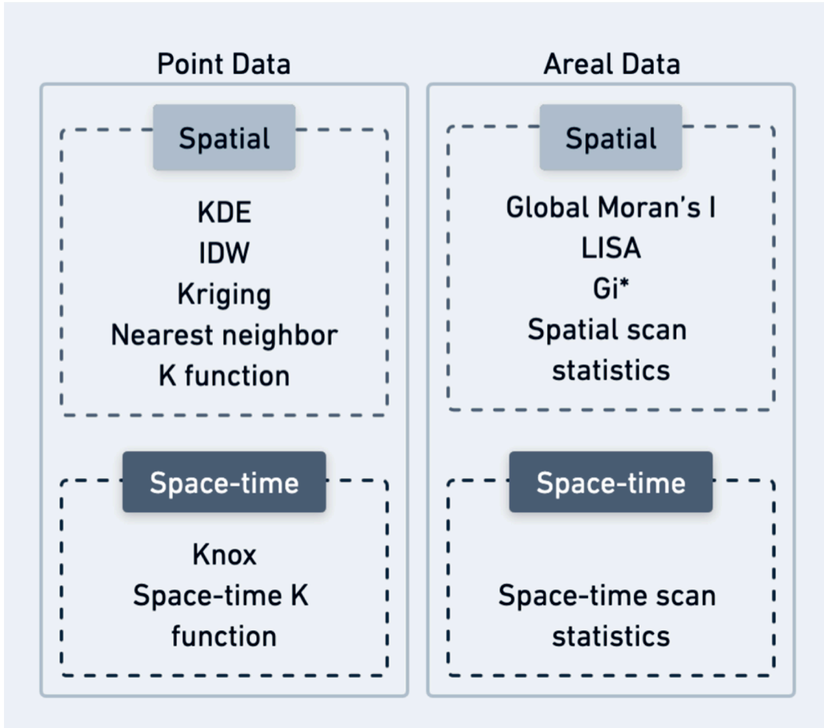

Fig. 1. Popular techniques for spatial and spatiotemporal cluster detection among point and areal data.

second-order techniques such as the nearest neighbor statistic and K-function can estimate clustering of the disease. The nearest neighbor statistic calculates the distance to the nearest neighbor, which is used to test whether closer incidents are randomly distributed or not (Delmelle, 2009). Two popular nearest neighbor techniques are the average nearest neighbor or *k*-nearest neighbors. The average nearest neighbor averages all nearest neighbor distances between each event and its nearest neighbor (also an event) and then returns the ratio of the observed mean distance to the expected mean distance for total events given in a random pattern. If this ratio is less than one, it indicates clustering. In the *k*-nearest neighbors (*k*NN) test, Cuzick and Edwards (1990) extended the nearest neighbor to *k* nearest neighbors of events, and this test can be used to detect spatial clustering of events with the consideration of the inhomogeneous populations.

The (Ripley's) *K*-function is another popular method to estimate second-order properties but assumes that no first-order effects exist in the spatial pattern (Pfeiffer et al., 2008; Ripley, 1977). In essence, the *K*-function is a count function of the number of other events within a circular search window around each event, and the window moves to the next event until all the events are visited (Hohl et al., 2017). The process is repeated for increasing radius values up to a maximum distance, coinciding with the two most distant point events. For evaluation of statistical significance, (random or population-based) simulations are generated by locating the same number of events in the study area, and the *K*-function is estimated for these simulations. Significance envelopes can be generated from these K-functions on simulated datasets; these are used to test whether the observed *K*-function falls inside, above, or below them (inside = randomness, below = dispersion, above = clustering).

When point events have a temporal signature (e.g., time of occurrence), several techniques described earlier can be extended to handle the temporal dimension, including the Knox test or the space-time *K* function. The Knox test is a pairing method to detect space-time clustering present in data points, based on the assumption that events' spatial and temporal features are independent of one another (Knox and Bartlett, 1964). The Knox test requires setting both a spatial and a temporal threshold distance, and it counts the number of pairs of events separated by critical space and time thresholds. It compares the observed and expected number of pairs of points using a Chi-Square test, where the expected number of pairs of points is calculated from simulated space-time point events. It is then possible to identify the space-time distance at which the Chi-square statistic is the greatest (note that multiple maxima can occur, suggesting clustering at different scales). The *K*-function has been extended by Diggle et al. (1995) to account for its temporal counterpart (e.g., Hohl et al., 2017). The mechanism of the space-time *K* function is, in essence, similar to the Knox test. Thus, the space-time *K* function can be considered a series of Knox tests with different critical distances. It measures the strength of clustering at different spatial and temporal scales which is informative for further space-time analysis.

1.3. Areal data

One key feature inherent to spatial data is spatial autocorrelation, which refers to the correlation of a variable of interest between two locations. Based on Tobler's first law of geography, correlation decreases with increasing spatial separation (Schabenberger and Gotway, 2017). Many tests can examine spatial autocorrelation, and these statistics are categorized into global and local statistics based on whether the returned result of clusters is nonspecific (global) or specific (local).

As to global statistics, these methods measure whether the pattern is clustered, dispersed, or random. The global Moran's *I* is one of the most popular spatial autocorrelation methods for areal data, it is based on both feature locations and feature values simultaneously, and the reported Moran's *I* value indicates whether events are statistically clustered (when $I > 0$). This test reflects the similarity among areas based on the assumption of the event distribution of the population at risk within the study area (Moran, 1950). The algorithm computes the mean and variance for the attribute of the variable being studied. Then, for each areal feature, it subtracts the mean, creating a deviation from the mean. Deviation values for all neighboring features (features within the specified distance band, for example) are multiplied together to create a cross-product. Thus, it provides a statistic (I) for each location with an assessment of significance (p-value). Second, it establishes a proportional relationship between the sum of the local statistics and a corresponding global statistic.

Local spatial autocorrelation methods are different from their global counterparts in that they are aimed at identifying the locations and extent of clusters (Pfeiffer et al., 2008). The local Moran's *I* test (LISA) is the local level version of the global Moran's I test, estimated by decomposing the Moran's *I* statistic geographically, returning local indicators of the spatial association (Anselin, 1995). The LISA statistic measures the strength of patterns among nearby geographic units, resulting in four different types of clusters. Low-Low clusters indicate low values surrounded by low values, High-High clusters indicate high values surrounded by high values, Low-High outliers indicate low values surrounded by high values, and High-Low outliers indicate high values surrounded by low values. These four categories can be used to lay out not only clusters of high values (e.g., High-High: the high infection rate area surrounded with other high infection rate areas) but also clusters with different surrounding situations (e.g., High-Low: the high infection rate area while surrounding areas have low infection rates).

Another local clustering detection method for aggregated data is the Getis-Ord Gi* statistic, which compares local estimates of spatial autocorrelation with global averages to detect clusters in the spatial data (Ord and Getis, 1995). It has been called the hot spot analysis since the test returns significant clusters of high (hot spot) and low (cold spot) values.

Another important clustering detection at the local level is the Kulldorff's spatial scan statistic. The scan statistic detects spatial clusters by scanning the data via a circular or elliptic window with the radius ranging from zero to a maximum value specified by the user (Kulldorff, 1997). A cluster is defined as a circle with a significant maximum likelihood ratio, and only events with centroids located within this circular are affiliated to this cluster. In addition, each cluster contains a relative risk (*RR*) value, which is the ratio of the estimated risk within the cluster to the calculated risk outside the cluster.

The Kulldorff's spatial scan statistic can be extended in time to account for the temporal dimension by replacing the scanning circular or elliptic window with a cylindrical window where the height represents the period of potential clusters (Kulldorff et al., 2005). The window with the maximum likelihood is the most likely cluster, that is the cluster least likely to be due by chance. Thus, this cylindrical window not only moves in space but also in time. In health surveillance, this statistic can be applied to both retrospective and prospective studies depending on the different focus on past patterns or current trends. Retrospective methods carry out analyses -such as hypothesis tests- for a fixed geographical region and a fixed study period to estimate the prevalence of diseases or compare diseases patterns in the different areas in one frame; in contrast, prospective methods use ongoing collected data and repeat analysis to detect a significant change in a timely manner (Sonesson and Bock, 2003).

To the best of our knowledge, no literature review has specifically addressed the use of space-time cluster detection techniques for infectious diseases. Several papers published more than ten years ago reviewed the techniques mentioned above, but none of them have investigated the use of space-time cluster detection methods in any systematic way. Robertson et al. (2010) presented a review of methods for space-time disease surveillance, with a focus on the role of contextual aspects including scale, scope, and other technical matters. Tsui et al. (2010) reviewed methods in spatiotemporal surveillance from a statistic perspective, including the Kulldorff's scan statistics, and pointed out the need to incorporate the estimation of outbreak coverage (spatial extent) and magnitude (risk level) into scan statistics in practice. Auchincloss et al. (2012) reviewed spatial methods in epidemiology and concluded that cluster analysis methods were among the most popular over the period of 2000 to 2010. They also point to the fact that although the majority of the studies were purely spatial, a few of studies did apply space-time clustering techniques to cancer and infectious diseases. Two recent review papers lead by Franch-Pardo (2020, 2021) focus on the use of spatial analysis and GIS for evaluating COVID-19 outbreaks. However, Franch-Pardo et al. (2020) only mentioned that some studies use prospective space-time statistics to "identify active and emerging COVID-19 groups"; in their second literature paper, Franch-Pardo et al. (2021) mentioned hotspots and clustering with a focus on purely spatial. Given the increasing availability of temporal information in infectious disease data, space-time analysis has become critical.

Thus, a systematic literature review, particularly of space-time cluster detection methods for infectious diseases, is needed given (1) the importance of time and not just geography when monitoring and surveilling the spread of infection, (2) the increasing awareness that spatial and spatio-temporal methods can provide additional information for public health. In addition, (3) numerous COVID-19 studies applied various detection methods to identify the space-time clusters, while how those methods were used has not yet been well discussed. This literature review can contribute to concluding what has been done before and what is still missing from literature and therefore shed light on improving currently used methods for this ongoing COVID-19 pandemic and future infectious diseases.

In Section 2, we describe the search and screening criteria used for this systematic review. In Section 3, we present results from our review; specifically, an overview of the topic in general and then in more descriptive ways, by disease types, geographic area, discipline, and data aggregation level. We particularly emphasize two very distinct approaches to space-time cluster detection, namely the temporal repetition of existing spatial methods and "true" space-time cluster detection methods. Section 4 provides a discussion of the meaningful findings of this research, directions for future research and concluding remarks.

2. Search and screening strategies

2.1. Search strategy

We conducted an electronic literature search for relevant articles from the PubMed, the Web of Science (WoS), and Scopus databases on August 27, 2022, articulated around four main queries (see Fig. 2). The first query included different types of infectious diseases but excluded non-human infectious studies. The second query attempted to incorporate articles that dealt with the spatial and temporal nature of contagious diseases (purely predictive studies, such as the ones using regression techniques, and which did not use clustering techniques, were excluded). In this query, we also excluded model-based approaches by adding "NOT" queries related to such approaches. The third query retained articles that focused on detecting spatial or space-time clusters and excluded genotype clustering papers. Finally, the fourth query further ruled out papers that were not relevant using specific keywords. Therefore, the first three queries are connected using the "AND" operator, while the last query uses "NOT" as a set of exclusion.

2.2. Screening and selection of criteria

Two authors individually screened the title and the abstract of the articles that matched our inclusion/exclusion criteria, in an effort to determine whether they needed to be fully reviewed. Articles were included if spatial and spatiotemporal analytical techniques were applied for the detection of clusters of infectious diseases for human populations (i.e., not animals). We also excluded articles using a model-based approach as it was beyond our scope of space-time cluster detection methods using statistical approaches. We used the Cohen's Kappa Statistic to evaluate the agreements between both reviewers for the screening, resulting in a *k* value of 0.77, which is considered a substantial agreement according to (Landis and Koch, 1977). For each manuscript where there was a disagreement, the two authors discussed the validity of the article; disagreement could stem from a lack of clarity

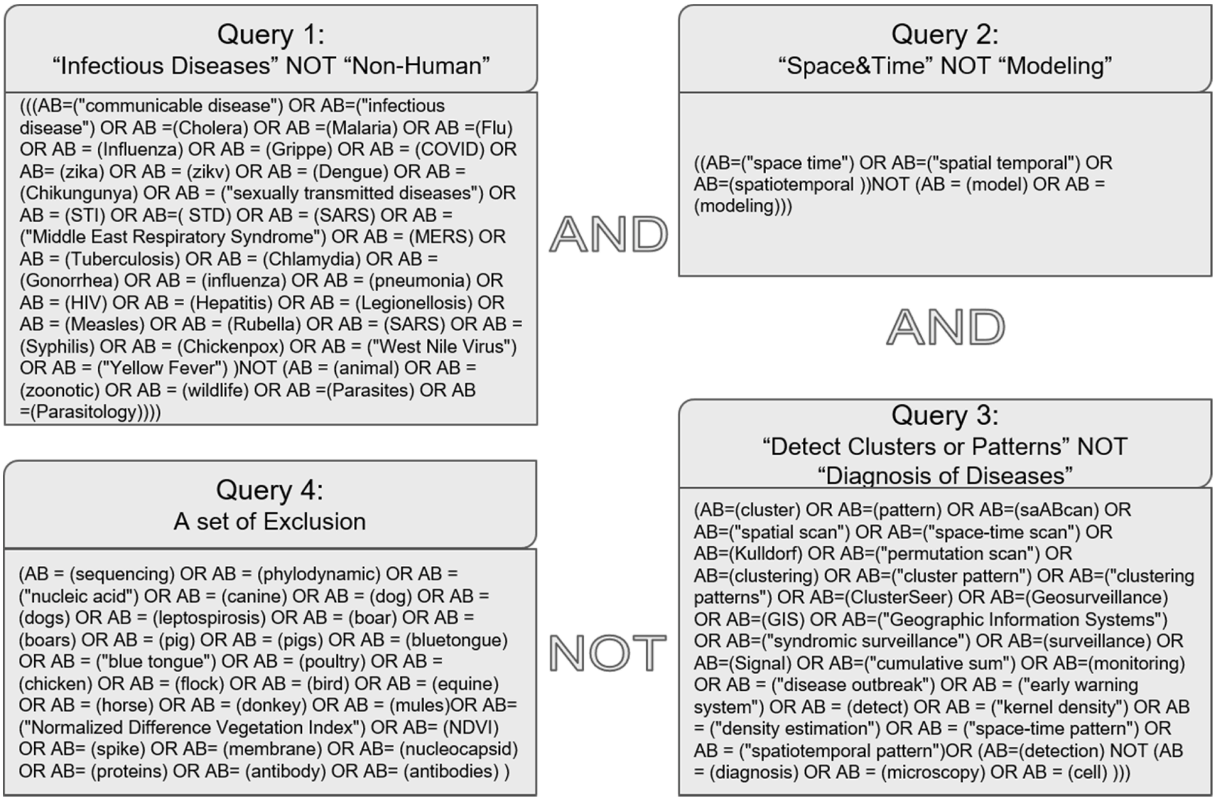

Fig. 2. Search queries (using the query examples for Web of Science).

and depth in the title or in the abstract, or that the abstract was misleading (for instance several papers discussed clustering of Dengue Fever, but the focus was on mosquito distribution). For each paper that had a disagreement, both authors reviewed the full text together, and arrived to a consensus. Only after both authors agreed on all manuscripts, the full text was reviewed together by both authors to confirm that those articles met the criteria described above.

3. Results

3.1. General summary

The searching and screening processes are summarized in Fig. 3. Using the four search criteria, a total of $n = 2{,}887$ articles published from 1974 to 2021 were identified from the PubMed, Web of Science, and Scopus databases ($n = 677$, $n = 540$, $n = 1{,}670$ respectively). After removing duplicates ($n = 811$) and review articles($n = 79$) returned by all the databases, that set of articles was reduced to $n = 1{,}996$ papers. We further excluded $n = 1{,}538$ articles during the screening phase; some articles were not related to raster studies ($n = 404$) or good sources (e.g., editor letters; $n = 45$), while others were identified as not relevant to the topic ($n = 1{,}089$). For the latter, articles were excluded when 1) there was no evidence of using space-time cluster analysis; 2) the study dealt with non-human infectious diseases; or 3) spatial regression or modeling methods were the main methods in the paper. The remaining articles ($n = 458$) were fully vetted for their eligibility. Of those, $n = 104$ were further flagged because they were not related to our search (e.g., review papers, regression-based papers, non-human diseases, etc.). Once this process was completed, a total of $n = 354$ articles spanning 44 years from 1977 to 2021 were included in this literature review with $n = 332$ articles (94%) that could be considered as application papers and $n = 22$ that were focused on methods. The list of articles can be found in the appendix.

3.2. Descriptive summary

This section provides a descriptive summary of our review according to disease types, study area, discipline, and data aggregation level. The number of articles published per year is summarized in Fig. 4, suggesting a marked increase every year, and especially so in 2020 and 2021 many COVID-19 studies (19 out of 53 in 2020 and 40 out of 68 in 2021) contributed to this increase. We illustrate our results for papers from 1977 forward since the number of articles prior to that date were not relevant to our criteria.

3.2.1. Disease types

Fig. 5 summarizes the frequency of articles by disease type (Dengue Fever, COVID-19, Tuberculosis, Malaria, Sexually Transmitted Infections, and other infectious diseases), while Fig. 6 reports the diseases which are most studied, year by year[1].

Overall, the first-largest number of articles was reported around vector-borne diseases (VBDs, $n = 142$). The most-reported VBD was Dengue Fever ($n = 83$), a disease caused by the Dengue virus, spreading from humans to humans through infected mosquitoes. Other VBDs with the same pathways, such as Chikungunya ($n = 2$) and Zika ($n = 2$), received less attention. A few studies reported the space-time prevalence of multiple VBDs in the same paper, including Dengue Fever, Chikungunya, and/or Zika. Malaria, another significant vector-borne disease caused by a parasite, received less attention ($n = 31$) than Dengue Fever. From Fig. 6, the number of studies related to Dengue Fever and malaria peaked in 2017 and 2018 but has decreased since then.

Articles related to airborne diseases ($n = 139$ including COVID-19, Tuberculosis, Influenza, and Respiratory Infections) formed the second

[1] Figs. 6 and 8 were started in 2004 due to the gap in the literature; only three articles selected before 2000 have limited effects on demonstrating the current trends.

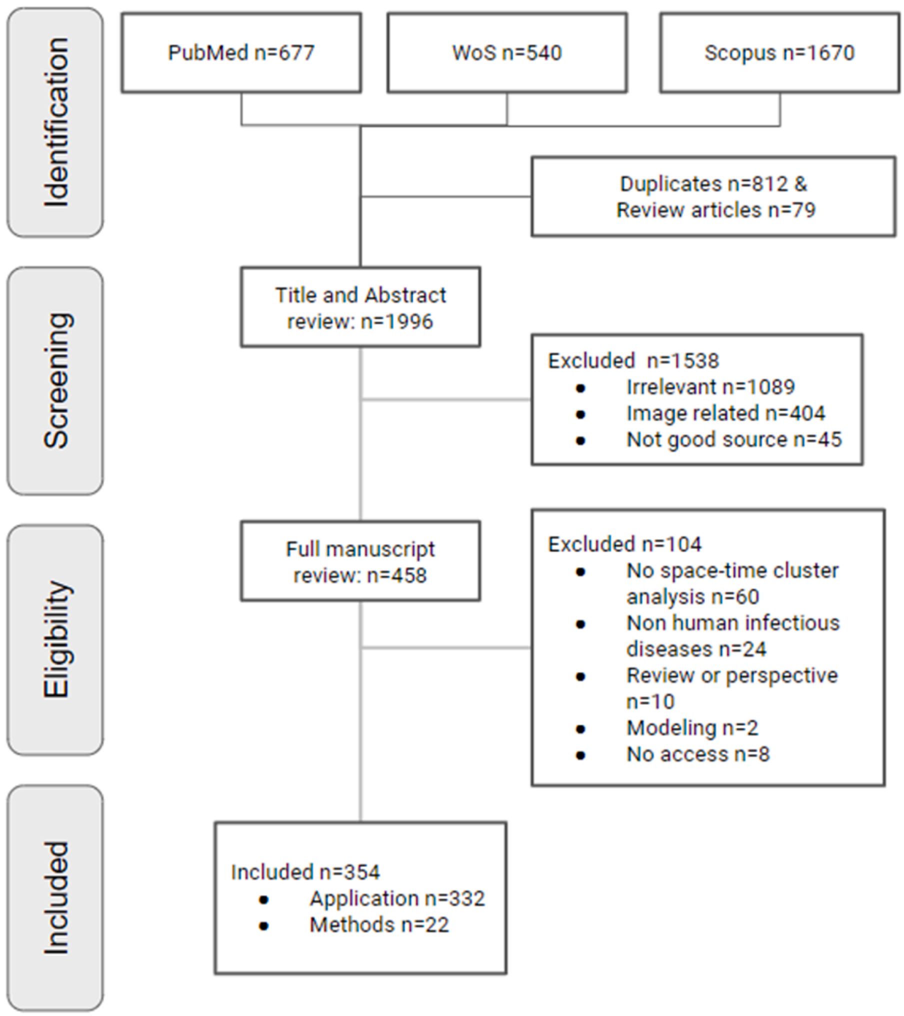

Fig. 3. Searching and screening results.

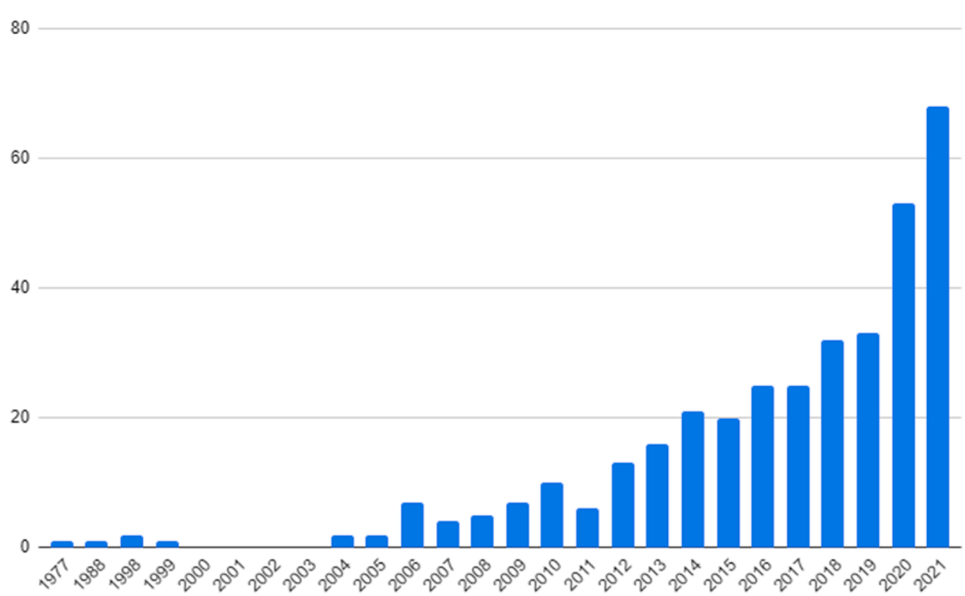

Fig. 4. Published articles per year (note that from 1978 to 1988 and 1988 to 1998 there were no papers reported).

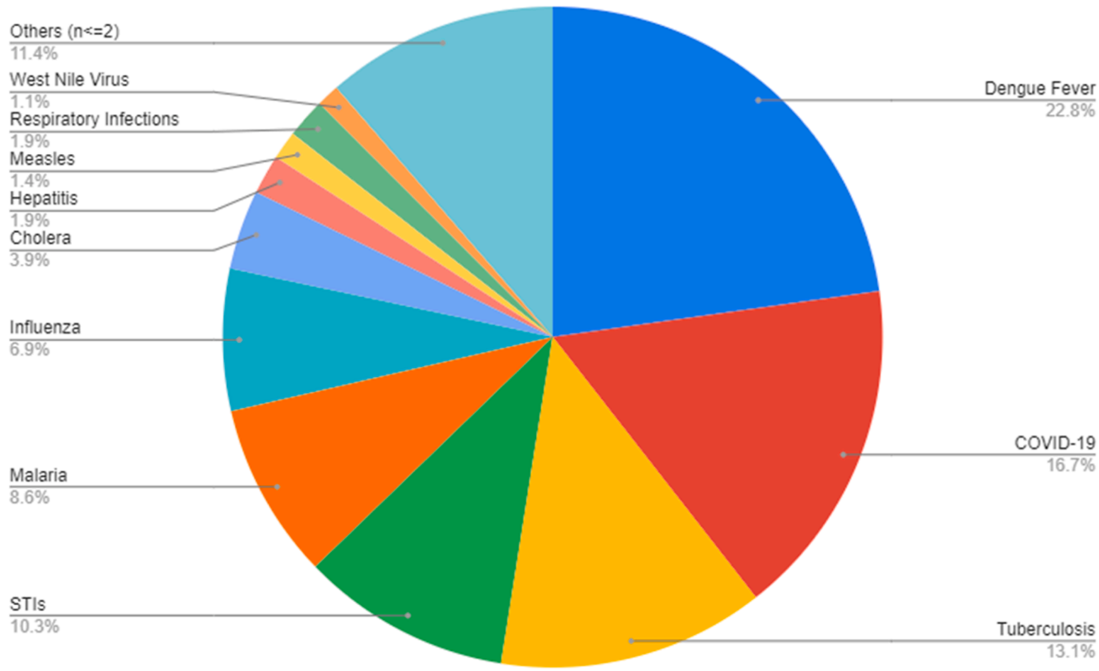

Fig. 5. Number and percentage of articles related to infectious diseases. When the number of articles for a particular disease was less than three, they were categorized as "others."

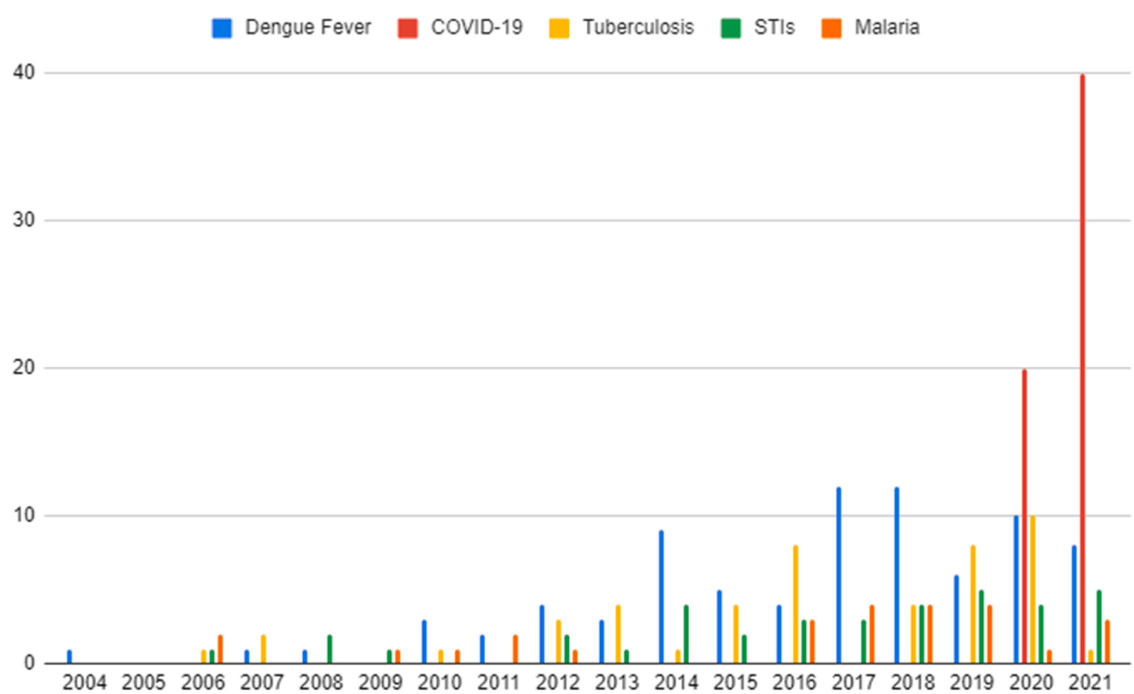

Fig. 6. The five most reported diseases.

most significant category. Two airborne diseases were ranked among the top five, namely COVID-19 ($n = 60$) and tuberculosis ($n = 47$). Influenza ($n = 25$) and Respiratory Infection ($n = 7$) were also ranked among the top ten of most documented infectious diseases. While the earliest articles on COVID-19 were published in 2020, the number of studies in 2021 was at least five times that for any other diseases (see Fig. 6).

STIs ($n = 37$) include infection from human immunodeficiency virus (HIV), Syphilis, Hepatitis B, Chlamydia, and others transmitted through sexual contact. Although the number of studies is less compared to the other two types of infectious diseases, it shows a steady increase since 2015, according to Fig. 6.

3.2.2. Study areas

Fig. 7 maps the number of case studies based on the countries or regions where the data originated from. The top 3 countries that have experienced the most studies were China ($n = 76$), US ($n = 35$), and Brazil ($n = 31$). Among studies in China, more than half ($n = 35$) are papers on airborne transmission including COVID-19 ($n = 18$), followed by VBD studies ($n = 17$). In the US, almost two third ($n = 22$) of the articles focused on airborne diseases, and half of these airborne disease

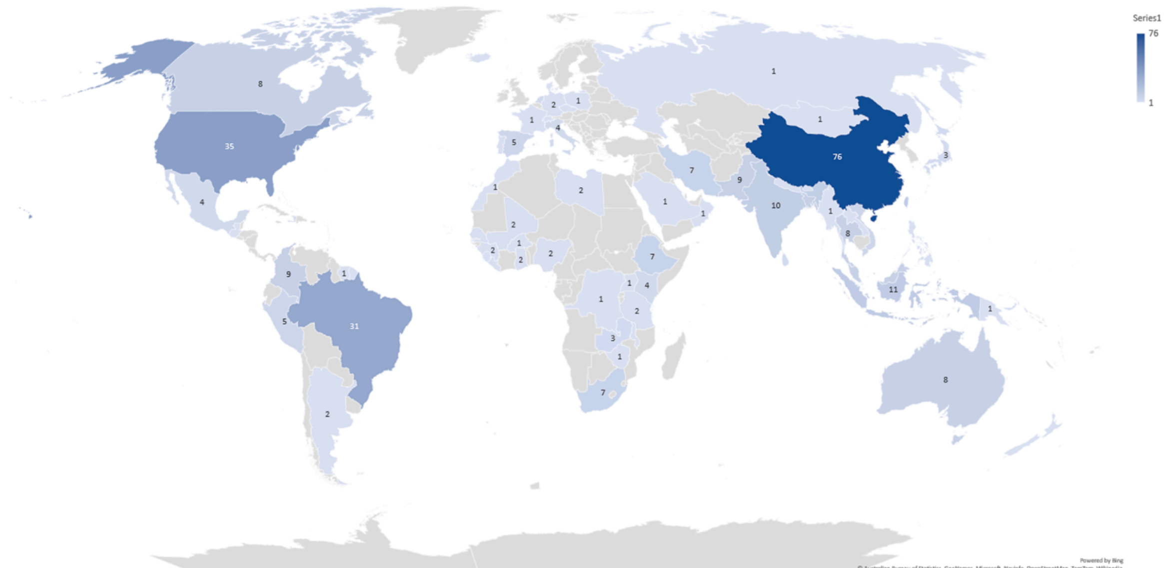

Fig. 7. Variation in the number of case studies by country and region.

studies were related to COVID-19 ($n = 12$). As to Brazil, most studies were also related to airborne diseases ($n = 13$ including $n = 8$ of COVID-19) and VBD ($n = 10$). It is worth noting that no studies were conducted in countries or regions located in central Africa, eastern Europe, and western Asia.

3.2.3. Discipline

We wanted to investigate the role of different disciplines for each article, because many studies are collaborative in nature. Manuscripts were categorized into different disciplines by extracting the first author's affiliation and the type of journal. The results showed that more than 60% of articles were published by authors from epidemiology or public health, while less than 30% were in geography or related background. In addition, a small percentage of studies were published from other disciplines, such as bioinformatics and mathematics.

We further compared the number of studies among different disciplines from 2004 to 2021 (see Fig. 8). The number of studies in epidemiology increased annually, while the number of studies originating in geography had a sharp increase since 2020. This growth is consistent with the rise of COVID-19 studies, as 26 studies related to COVID-19 (44% of all COVID-19 studies) were published from within geography. More geographers have been involved in studies of COVID-19 and other infectious diseases, particularly in detecting spatial and temporal patterns.

3.2.4. Data types

About a fourth of studies used point data, while others used aggregated data. One important reason for the dominance of areal data is that such data types are more readily available, partly because of an effort to protect patient confidentiality by preventing disclosure of a patient's

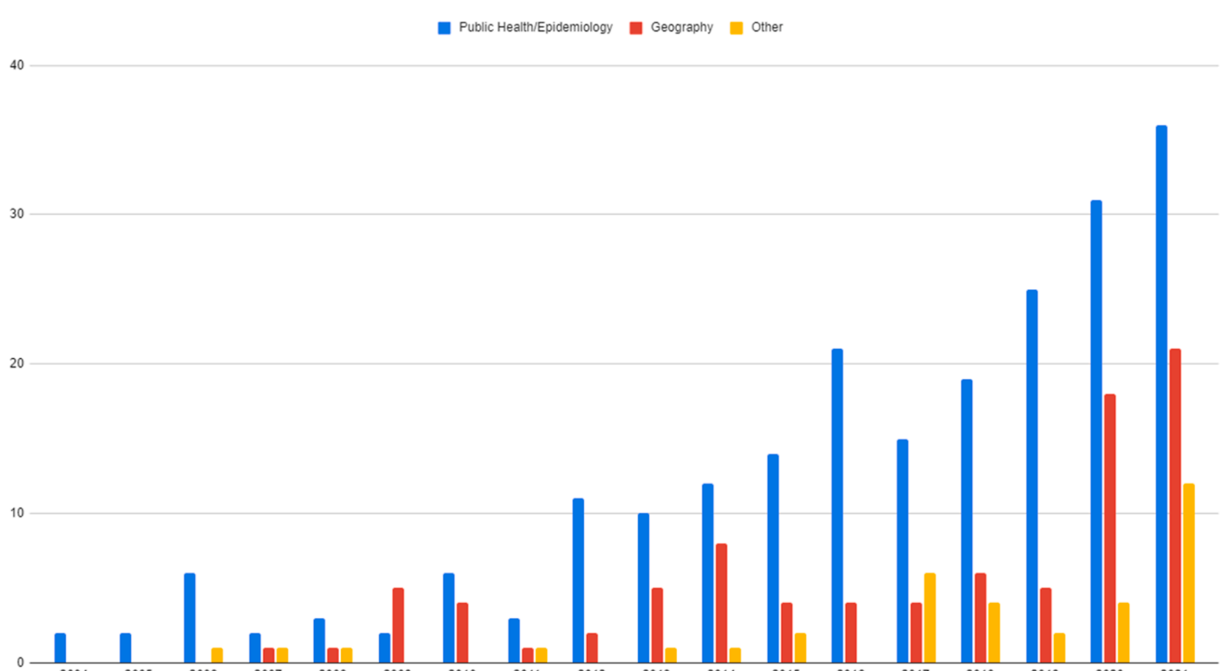

Fig. 8. Studies by discipline and year.

identity (Goovaerts, 2009). In addition, for those studies using aggregated data, the geographic scale was either at the county (27.4%) or local level (57.4%), while only 11.6% of studies were at state (10%) or country (2%) level.

3.3. Space-time cluster detection methods

This section summarizes the most common techniques of space-time cluster detection methods, including temporal repetition of spatial methods and the exact space-time methods, found in the literature (see Fig. 9 for a summary of the frequencies). The most popular technique was the Kulldorff's space-time scan statistics (n = 205 articles). This number is almost twice as high when compared to studies that used the (Global) Moran's I method (n = 79), the second most popular technique. The third and fourth most popular techniques are the Local Moran's I (n = 65) and the Local Gi* (shortened for Getis-Ord, n = 44). It is worth noting that several articles compared and contrasted the results of more than one technique in the same paper. The purely spatial scan statistics was also very popular, (n = 32).

We distinguish two approaches to perform space-time detection: either a temporal repetition of an existing spatial method or a true, space-time approach that explicitly considers space and time. Table 1 shows the number of studies using those two approaches to analyze space-time patterns in infectious diseases and the data type used, suggesting that more studies used true space-time methods instead of temporal repetition of spatial methods.

3.3.1. Temporal repetition of spatial methods

Temporal repetition of spatial methods identifies space-time patterns of diseases by repeating a spatial method for different temporal intervals. Generally, this approach is based on spatial smoothing and interpolation for point data and purely spatial cluster detection for both point data and aggregated data.

3.3.1.1. Spatial smoothing and interpolation. A total of 23 studies used interpolation and smoothing methods, including KDE, to reconstruct the spatial variation of the patterns across multiple timespans. Several studies applied interpolation methods to detect space-time clusters by generating continuous estimated surfaces from point data at multiple time intervals (See McIntosh et al., 2018; Singh and Chaturvedi, 2021). For example, de Azevedo et al. (2020) used KDE to create yearly density maps of dengue outbreaks in Brazil from 2000 to 2018 and to identify outbreaks. In an article by Pardhan-Ali et al. (2012), the authors used ordinary kriging to generate a relative risk map of notifiable gastrointestinal illness in the Northwest Territories of Canada, while relative risks were estimated using the results of the spatial scan test.

Table 1
The number of studies using two types of space-time detection techniques.

	Temporal repetition of spatial methods	Space-time methods	Both	Total
Point	23	48	12	83
Aggregated	87	134	55	276
Total	110	182	67	359 (n = 4 using both point and aggregated data)

Although temporal extensions for these methods have been proposed in the literature (such as STKDE, space-time interpolation), they are computationally demanding, and despite one notable exception (Hohl et al., 2022), these approaches do not take population change into account during the period under consideration. The results of the STKDE are best visualized in a three-dimensional framework, however this is computationally demanding and can be cognitively challenging. Therefore, a simple repetition of the same method for different time ranges is generally preferred to understand the temporal variation of a disease (see de Azevedo et al., 2020; Sifuna et al., 2018).

3.3.1.2. Spatial clustering detection methods. Among that implemented a temporal repetition of spatial methods, most of them applied spatial cluster detection (e.g., LISA) instead of spatial smoothing or interpolation. Few studies applied the nearest neighbor techniques (n = 3) to detect the scale at which clusters were dominant. Some papers used kNN to generate spatially smoothed dengue incidence maps, but not for the cluster detection (e.g., Acharya et al., 2016).

From the literature review, n = 79 articles implemented the global

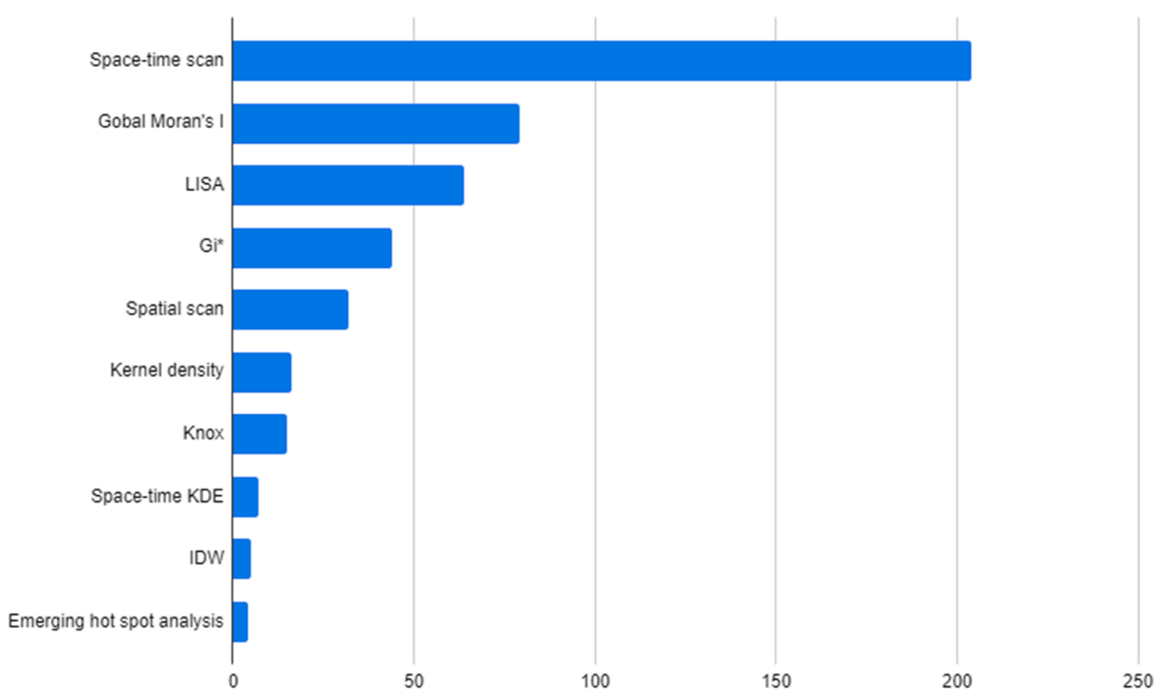

Fig. 9. Frequency of studies by statistical technique.

Moran's *I* statistic to inform on the presence of areal clustering. For example, Lippi et al. (2020) used the Moran's *I* statistic on annual aggregated dengue cases at the level of the health districts and polyclinic administrative catchment (PAC) areas in Barbados from 2013 to 2016. Similarly, Yu et al. (2020) conducted the Moran's *I* statistic on annual aggregated rates of pulmonary tuberculosis (PTB) in counties of Chongqing, one of the biggest cities in China, from 2011 to 2018, suggesting significant clusters each year.

Several studies ($n = 100$) used either the LISA or Gi* to detect local clusters. In Lippi et al. (2020), the LISA algorithm identified the dynamic patterns of both high and low clusters of Dengue Fever, revealing a shift in the spatial patterns of clusters at a local level (PAC level). Yu et al. (2020) also conducted the local Gi* statistics for PTB at the county level in Chongqing, China, reporting yearly statistically significant clusters. As Gi* compares local estimates of spatial autocorrelation with global averages to detect hotspots, this method may not be suitable for small sample sizes, implying high levels of global autocorrelation (Getis and Ord, 1996).

Instead of considering spatial autocorrelation, Kulldorff's spatial scan statistic ($n = 32$) is applied to detect the presence of local clusters of infectious diseases (see Rocheleau et al., 2020). Many studies used this scan statistic to estimate the relative risk (RR) or risk ratio, which can help to compare this risk of infection among areas (see Rejeki et al., 2019; Sloan et al., 2020).

3.3.2. Space-time methods

Unlike temporal repetition of spatial approaches, space-time methods include both the spatial and the temporal dimensions in statistical tests. The first study from our search applied a trend-surface analysis by using the time as the third vibrable to generalize cubic surfaces to map the space-time distribution of an epidemic in a Brazilian city in 1956 (Angulo et al., 1977). Some of the most popular techniques in this category from the literature are introduced as follows.

A few studies applied the Knox test ($n = 15$), and in ten articles, Knox was applied to Dengue Fever (e.g., Tran et al., 2004; Vazquez-Prokopec et al., 2010; Wen et al., 2012). One possible reason is the clear space-time transmission among infected mosquitoes in dengue disease, making it easy to set the critical threshold of space and time distance. However, because it is a global test, the Knox test is not good at visualizing cluster information on the map. Therefore, some papers have relied on additional visualization methods to display these space-time patterns. For instance, Rotela et al. (2017) used the Knox test for spatial-temporal analysis and KDE with a 300-meter radius to show the density of Dengue cases in Cordoba, a city in Argentina.

Only five studies have used the space-time *K* function, primarily due to heavy computational requirements. Hohl et al. (2016) for instance calculated spatial and temporal bandwidths using the space-time *K* function on daily Dengue Fever cases in Cali, Colombia from 2010 to 2011. They used these results (different bandwidths) as inputs to estimate the space-time kernel density, and ultimately visualized results in a three-dimensional space-time cube.

According to our review, the space-time scan statistic is the most popular technique to detect clusters for aggregated data ($n = 204$). In a recent study, Hohl et al. (2020) used a prospective space-time scan statistic to estimate clusters of COVID-19 in the US at the county level. One possible reason for the popularity of the scan statistic lies in its ability to incorporate multiple covariates (e.g., Whiteman et al., 2019), including population. For example, in infectious diseases, the population is a critical variable that needs to be considered in the analysis, while many other pattern detection methods cannot include this covariate directly. Also, some studies focus on spatial although space-time methods were used. For example, Tadesse et al. (2013) concluded that both purely spatial and space-time scan tests detected similar and significant high-risk clusters of smear-positive TB cases in a district of Ethiopia, but no temporal information on clusters was provided in the study. In another article, even though Gurjav et al. (2015) claimed to use the retrospective space-time scan test and detect three TB clusters in Mongolia from 2006 to 2012, no temporal characteristics of clusters were provided.

Space-time clusters can be detected by either space-time methods (e.g., space-time scan test) or repetition of purely spatial methods (e.g., LISA). Although both methods returns geographic units considered to be in a cluster, the results are generally different. Fuentes-Vallejo (2017) implemented both Gi* and the space-time scan statistic to compare the different sensitivity of local clusters from these two methods. Their results show that spatial clusters (using Gi* statistic for each year) and space-time clusters from the scan test were located in different regions, although some results were overlapping. They claimed that this different spatial and spatiotemporal clusters distribution is possibly due to different territorial dynamics.

Overall, few studies applied cluster detection approach into a web environment. Reinhardt et al. (2008) launched an online GIS, EpiScanGIS, integrated SaTScan to monitor the invasive meningococcal disease in German, while the system was not in function now. Beside EpiScanGIS, only five recent studies on COVID-19 published in 2020 and 2021 deployed their detection approach in a web-based application. Four of them used the prospective space-time scan statistics, while one used another algorithm (a modified space-time density-based spatial clustering of applications with noise). Two studies are similar in that their online systems map daily clusters as an animation based on daily COVID-19 cases (Hohl et al., 2020; Rosillo et al., 2021), and another study, instead of mapping clusters, animated daily relative risk and cluster frequency results (Lan et al., 2021). Two other studies (De Ridder et al., 2021; Güemes et al., 2021) detected space-time clusters of COVID-19 symptoms by collecting symptoms from users. But in none of these five articles were users able to conduct customized space-time analysis, indicating that the results of the clustering were done beforehand, and subsequently uploaded on the web environment.

Although most studies use traditional visualization techniques (e.g., small multiples) to map space-time clusters, a few studies ($n = 13$) introduced novel methods to portray space-time clusters. The most popular method ($n = 11$) is to map the dynamics of space-time clusters along the third dimension that represented time. Among those studies, six of them used three-dimensional (3D) figures to represent cluster areas along with the time axis (e.g., see Desjardins et al., 2018), and five of them delineate space-time clusters by rendering values into different colors or/and transparency in the 3D volume, which have no well-defined boundary (e.g., see Kuo et al., 2018). Other than 3D methods, three studies used ring maps (Tang et al., 2019), calendar-based visualization (Wu et al., 2021), and bivariate and spike maps as different visualization of space-time clusters (Lan et al., 2021).

3.4. COVID-19 studies

From our systematic review, several studies were applied to the analysis of COVID19 outbreaks ($n = 60$). The number of COVID-19 articles that used a space-time algorithm ($n = 34$) was nearly the same as the papers that used a repetition of a spatial clustering algorithm ($n = 32$); the latter approach is very salient to distinguish spatial patterns from one time period to another, while six studies use both algorithms. Studies conducted by epidemiologists ($n = 25$) and geographers ($n = 26$) were close, and the remaining studies ($n = 9$) were led by researchers from other disciplines. Most studies used aggregated data ($n = 51$), and 48 of those studies were conducted at the county or more local level.

4. Conclusion and discussion

This article presents a systematic literature review that reflects recent trends in space-time cluster detection for infectious diseases. This is the first study to undertake a systematic review on this topic, and we hope that this research will contribute to a deeper understanding of the use of space-time analysis methods in the context of infectious diseases.

We searched and selected 354 articles from PubMed, Web of Science, and Scopus databases. The number of articles exhibited a continuous increase from 2004 to 2021, and nearly doubled from 2019 to 2020 and 2021 due to the emergence of research related to COVID-19. Most of the articles were application type papers that use spatial and spatiotemporal techniques to detect space-time clusters for infectious diseases. Very few studies attempted to publish their results over a web-based interfaces, and the visualization results were for the most part two dimensional. Overall, research on airborne diseases were dominant, followed by vector-borne diseases. Most studies were conducted in China, U.S., and Brazil. In 2020 and 2021, several geographers published articles related to COVID-19, contributing to 42% of all COVID-19 studies for the mapping and detecting of space-time clusters.

As to techniques related to space-time cluster detection, we note the following trends. First, most studies use aggregated data instead of point data. Second, there was a greater number of studies using “true” space-time detection algorithms as compared to papers only using temporal repetitions of spatial methods. Third, the most popular methods were the space-time scan statistics, the global Moran’s *I*, and the LISA statistic.

Several research gaps can be identified in our current understanding of this field. From the literature review, few online applications or platforms have implemented spatial or space-time analytical techniques to identify space-time clusters of infectious diseases. Only five online applications incorporated those techniques, and all of them are COVID-19 related with limited functionality. Some commercial GIS platforms, such as ArcGIS Online, have implemented specific spatial cluster detection techniques (e.g., LISA) online. However, these platforms require credits to access these resources. Further, none of these applications can conduct an analysis that would account for both spatial and temporal dimensions simultaneously. In other words, users can only conduct a repetition of these purely spatial methods by performing the same analysis for different temporal intervals. Other issues that may prevent the deployment of web-based platforms lie in their scalability, computation, and high-level programming skills to develop such systems.

Another critical question requiring more attention is how to visualize space-time clusters (only 13 articles deliberately discussed this issue). Using the space-time cube as a framework, the third dimension can be used to visualize the dynamics of space-time clusters better, potentially uncovering hidden space-time patterns. New technologies for web-based data visualization, such as WebGL and D3.js, can be used to visualize space-time patterns on the internet.

As far as the literature review is concerned, no study implemented space-time analysis and visualization into one web platform. For space-time cluster detection of infectious diseases, those two components are complementary. A good analysis of space-time clusters could undermine the conclusion without proper geovisualization and vice versa. Tight-coupled systems that can handle both analysis and geovisualization are greatly needed.

We also underline several methodological concerns in applying such methods, promising to improve future studies. It is worth noting that repeating a spatial clustering algorithm across time instead of using a true space-time cluster detection technique may cause an increase of type I and type II errors (false positive and false negative, respectively), and almost half of the studies found in the literature review used a temporal repetition of a spatial clustering method alone or together with “true” space-time detection methods. In addition, the temporal range used in these approaches is usually arbitrary (e.g., week, month, trimester) based on the dataset or personal experience; analysis conducted at different scales can further exacerbate type I and type II errors.

As mentioned earlier, space-time statistics can be applied both retrospectively and prospectively, but each approach answers different research questions. Retrospective cluster detection conducts the analysis once and identify all existed clusters during the whole study time, while the prospective method conducts the analysis on multiple time interval (e.g., daily, weekly, or yearly) to detect ‘alive clusters’ on each end date of that time interval during the study period. The retrospective method scans to detect clusters from the end of the study to the beginning, while the prospective one moves reversely through time. Mainly, retrospective methods detect patterns for a fixed dataset, while prospective methods adapt the results considering both newly available and past data. Therefore, prospective methods are more appropriate to promptly detect the dynamics of space-time clusters, especially during an outbreak that needs a rapid response. As to COVID-19, retrospective methods can determine whether it will become a seasonal recurrent disease like flu when cases are recorded for more than one year, while prospective methods can closely monitor the change in the current situation.

Some studies also overlooked the importance of interpreting the temporal information of space-time clusters. In some articles, the temporal extent of space-time clusters was not presented, although space-time scan tests were used. Without an adequate examination of the temporal characteristic of space-time clusters, their ability to offer additional insights in the temporal dimension vanishes.

Since 2011, multidisciplinary collaboration has steadily increased. In addition, the collaboration among the academy, government, and research centers represented nearly 50% of all the publications. Both results suggest that investigating the presence of infectious diseases is best tackled by a holistic team of researchers. Thus, it is also essential to incorporate multiple levels of collaboration across academics, health agencies, and other organizations. This kind of collaboration will offer theoretical evidence to support the implementation of health policies and practical experiences to guide research design and evaluation.

Several issues warrant further investigations. First, most articles in the literature did not account for the potential effects of scale. For instance, clusters identified from cases reported at the postal code level (e.g., ZIP in the US) may not be the same as if data were reported at the county level. We argue that the comparison among multiple scales for the same study region (e.g., county and census tract levels) could provide additional insights into the mechanism of the disease under study. Second, research should more explicitly discuss the temporal dimension in cluster detection, because it can reflect the cyclicity and dynamic nature of a disease. Both spatial and temporal dimensions are equally essential for the monitoring of infectious diseases. Third, more research is needed to compare the validity of clusters found from a repetitive spatial method, or from a true space-time clustering algorithm as it can affect the risk of false alarms. Fourth, other clustering techniques such as wombling (Hossain and Lawson, 2005; Hossain and Lawson, 2010; Lu and Carlin, 2005; Monir Hossain and Lawson, 2006) did -which identifies various levels of cluster boundaries- are promising, but are rarely used for temporal processes, nor in infectious diseases. There is a potential to extend these approaches in time. Fifth, our literature review did not explicitly search for papers using Space-Time Bayesian modeling; in fact, most of the algorithms discussed in our paper are to describe and identify space-time clusters; as such this topic falls outside of the scope of this research. Sixth, we found several studies that used the space-time K function to detect space-time clustering from events, but the inhomogeneous K function (Baddeley et al., 2000) has rarely been discussed in space and time (a notable exception is (Hohl et al., 2022)); this is partly due to the difficulty to have temporally varying information on the population itself, unless the study covers a large period of time, and fine-grained population count is available. Finally, researchers should facilitate the replication of their study, either by publishing their data and developing web-based visualization solutions.

5. The PRISMA protocol summary

- We describe the rationale of this review on page 6, lines 35-38.

“To the best of our knowledge, no literature review has specifically addressed the use of space-time cluster detection techniques for infectious diseases. Several papers published more than ten years ago reviewed the techniques mentioned above, but none of them have investigated the use of space-time cluster detection methods in any systematic way."

- We provide the statement of the objectives the review addresses on page 7, lines 8-16.

"Thus, a systematic literature review, particularly of space-time cluster detection methods for infectious diseases, is needed given (1) the importance of time and not just geography when monitoring and surveilling the spread of infection, (2) the increasing awareness that spatial and spatio-temporal methods can provide additional information for public health. In addition, (3) numerous COVID-19 studies applied various detection methods to identify the space-time clusters, while how those methods were used has not yet been well discussed. This literature review can contribute to concluding what has been done before and what is still missing from literature and therefore shed light on improving currently used methods for this ongoing COVID-19 pandemic and future infectious diseases."

- We specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses on page 7, line28-36.

"The first query included different types of infectious diseases but excluded non-human infectious studies. The second query attempted to incorporate articles that dealt with the spatial and temporal nature of contagious diseases (purely predictive studies, such as the ones using regression techniques, and which did not use clustering techniques, were excluded). The third query retained articles that focused on detecting spatial or space-time clusters and excluded genotype clustering papers. Finally, the fourth query further ruled out papers that were not relevant using specific keywords. Therefore, the first three queries are connected using the "AND" operator, while the last query uses "NOT" as a set of exclusion."

- We specify all databases searched to identify studies and the date when each source was last searched or consulted on page 6, lines 43-44.

"We conducted an electronic literature search for relevant articles from the PubMed, the Web of Science (WoS), and Scopus databases on August 27, 2022, articulated around four main queries (see Fig. 2)."

- We present the full search strategies for all databases including any filters and limits used on page 8, Fig. 2.
- We specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record, on page 7-8, lines 38-7.

"Two authors individually screened the title and the abstract of the articles that matched our inclusion/exclusion criteria, in an effort to determine whether they needed to be fully reviewed. Articles were included if spatial and spatiotemporal analytical techniques were applied for the detection of clusters of infectious diseases for human populations (i.e., not animals)...For each manuscript where there was a disagreement, the two authors discussed the validity of the article; disagreement could stem from a lack of clarity and depth in the title or in the abstract, or that the abstract was misleading (for instance several papers discussed clustering of Dengue Fever, but the focus was on mosquito distribution). For each paper that had a disagreement, both authors reviewed the full text together, and arrived to a consensus. Only after both authors agreed on all manuscripts, the full text was reviewed together by both authors to confirm that those articles met the criteria described above."

- We describe the method that we conducted to assess the robustness of the synthesized results on page 7, lines 41-43.

"We used the Cohen's Kappa Statistic to evaluate the agreements between both reviewers for the screening, resulting in a k value of 0.77, which is considered a substantial agreement according to (Landis and Koch, 1977)."

- We describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, using a flow diagram (Fig. 3) on page 9.
- We summarize statistics for each group using structured figures (Figs. 4-9) or a table (Table 1).
- We provide a general interpretation of the results in the Discussion section.

Declaration of Competing Interest

The authors have no conflicts of interest.

Data Availability

Selected articles are available as appendix.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.sste.2022.100563.

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